

# Common Denominator

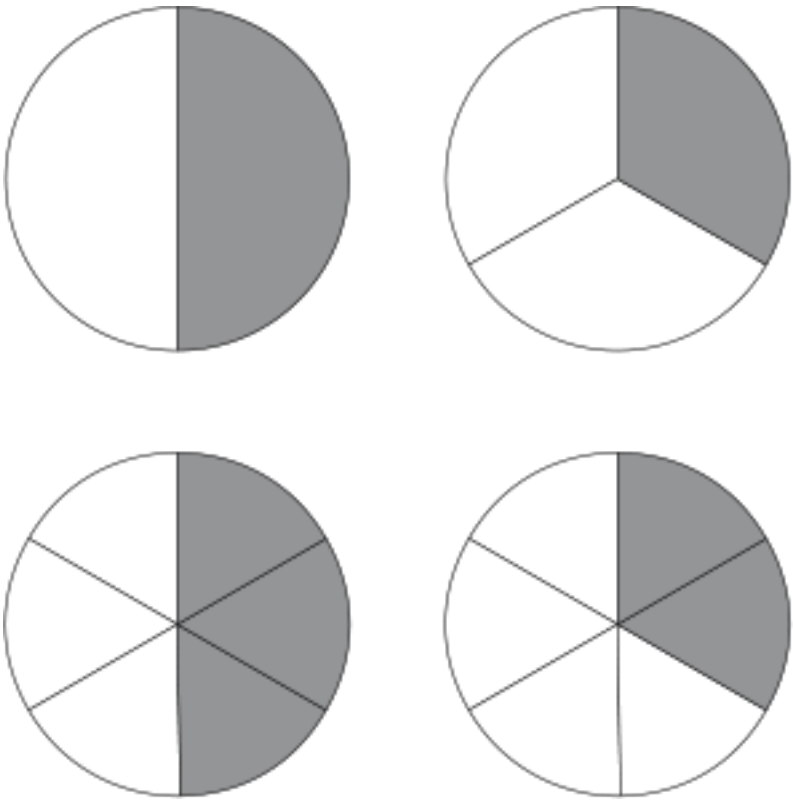
You can add and subtract fractions only if they have the same denominators.

Imagine trying to add  $\frac{1}{2}$  and  $\frac{1}{3}$ . The circles are cut into sections of different sizes, so there isn't a way to combine them.

To add them, look for a way to cut both circles into sections of the same size.

Here the circles are cut into sixths.  $\frac{1}{2}$  becomes  $\frac{3}{6}$ .  $\frac{1}{3}$  becomes  $\frac{2}{6}$ .

Now you can add:  $\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$



In this lesson, you will practice finding a common denominator and writing equivalent fractions.

### Example 1

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You can always find a common denominator by multiplying the denominators.

Find a common denominator for  $\frac{3}{4}$  and  $\frac{1}{6}$ , and write equivalent fractions.

**Step 1**  
Multiply the denominators.  
 $4 \times 6 = 24$

**Step 2**  
Write equivalent fractions with the denominator 24.

$$\frac{3}{4} \times \frac{6}{6} = \frac{18}{24}$$
$$\frac{1}{6} \times \frac{4}{4} = \frac{4}{24}$$

Multiplying the denominators isn't the best way to find a common denominator, but it will always work.

The best way to find a common denominator is to find the **least common multiple (LCM)** for both denominators.

### Example 2

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List the first few multiples for each denominator. Then look for the lowest number that both lists have in common.

Find a common denominator for  $\frac{3}{4}$  and  $\frac{1}{6}$ , and write equivalent fractions.

**Step 1**  
List a few multiples for both numbers.  
**4:** 4, 8, 12, 16, 20, 24, ...  
**6:** 6, 12, 18, 24, ...

**Step 2**  
Find the least common multiple.  
The lowest number in both lists is **12**.

**Step 3**  
Write equivalent fractions with the denominator 12.

$$\frac{3}{4} \times \frac{3}{3} = \frac{9}{12}$$
$$\frac{1}{6} \times \frac{2}{2} = \frac{2}{12}$$

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